

BanglaSpeechFM: Towards a Foundation Model for Bangla Speech

Anonymous ACL submission

Abstract

Self-supervised learning has greatly improved speech models, but low-resource languages like Bengali still rely heavily on large, expensive multilingual models. These large models are difficult to run and often miss the specific phonetic details of Bengali. In this paper, we introduce a Continuous Latent Autoencoder (CLAE), which is an efficient 23.8M-parameter speech foundation model built from scratch for Bengali speech. We trained it on public datasets using a new self-supervised approach that combines mel-spectrogram reconstruction, Joint-Embedding Predictive Architecture (JEPA), and Variance-Invariance-Covariance Regularization (VISReg). This allows our model to create strong representations at a 12.5 Hz latent sequence rate. Even with a small 15.29M-parameter frozen feature extractor, our model performs just as well as, and sometimes better than, much larger state-of-the-art speech models. Our results show that training a small foundation model from scratch is a practical and effective way to handle low-resource speech tasks. This provides an accessible and resource-friendly option for Bengali speech research and helps set the stage for future work in similar languages.

1 Introduction

Speech representation learning has seen remarkable progress with the advent of self-supervised learning (SSL) techniques, enabling the development of foundation models capable of understanding and processing human speech. Traditional SSL approaches in speech have often relied on reconstructing raw audio or spectrograms, which can be computationally demanding and prone to capturing low-level noise rather than high-level semantics. Recently, paradigms such as the Joint-Embedding Predictive Architecture (JEPA) have emerged as highly effective alternatives. By masking regions in high-level feature spaces and predicting latent representations rather than raw pixels or waveforms,

JEPA encourages models to learn rich, contextual semantics. While state-of-the-art (SOTA) speech models have achieved unprecedented success in high-resource languages by leveraging such techniques, low-resource languages like Bengali often suffer from a lack of dedicated, efficient models built from scratch.

Most existing works in Bengali speech AI rely on fine-tuning massive, pre-trained multilingual models, which are computationally expensive and not optimized specifically for the linguistic nuances of Bengali. Furthermore, the immense size of these SOTA models limits their deployability in resource-constrained environments. This research addresses this gap by developing an AI/ML speech foundation model specifically for Bengali, trained entirely from scratch using self-supervised learning on publicly available datasets. By doing so, we demonstrate that training a compact, dedicated model from scratch is not only viable but can also yield highly effective representations for Bengali speech.

Despite the growing interest in speech technologies, there is a significant scarcity of foundational speech models built from scratch for the Bengali language. Current approaches predominantly involve fine-tuning extremely large SOTA models trained on other languages or multilingual corpora. This reliance on massive pre-trained models presents two major challenges: first, the computational cost and memory footprint of these models make them impractical for widespread deployment or research with limited resources; second, they often lack specialized adaptation to the unique phonetic and phonological characteristics of Bengali. There are currently no researchers who have successfully developed a small-scale, highly efficient speech foundation model for Bengali from scratch using self-supervised learning on public datasets. Therefore, there is a critical need to investigate whether a significantly smaller, dedicated model

can be trained from scratch to achieve competitive or superior performance in specific metrics compared to existing heavy-weight SOTA models.

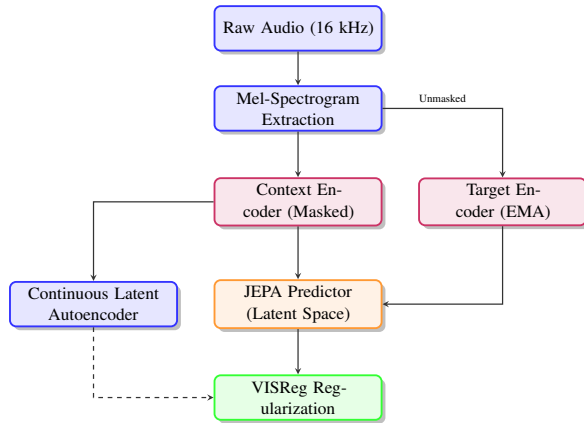


Figure 1: Proposed self-supervised CLAE framework for Bengali speech representation, integrating a Joint-Embedding Predictive Architecture (JEPA) and VISReg regularization.

To address the aforementioned challenges, our work introduces a novel learning objective that optimally combines mel-spectrogram reconstruction, JEPA-style latent space prediction, and Variance-Invariance-Covariance Regularization (VISReg), as visually outlined in Figure 1. VISReg ensures that the learned representations maintain high variance (preventing informational collapse), remain invariant to certain augmentations or masked noise, and decorrelate covariance among features, resulting in robust and disentangled speech embeddings.

The primary objective of our work is to design, train, and evaluate a Continuous Latent Autoencoder (CLAE) for Bengali speech representation learning. By developing a compact speech foundation model from scratch using self-supervised learning techniques, we aim to provide a highly efficient alternative to fine-tuning large SOTA models. The specific objectives of this research are summarized as follows:

- To develop a lightweight speech foundation model (a 23.8M-parameter continuous-latent autoencoder) specifically tailored for 16 kHz Bengali speech using publicly available datasets.
- To demonstrate that training a model from scratch using self-supervised learning is a viable and highly effective pathway for Bengali speech research, eliminating the strict dependency on fine-tuning massive multilingual

models.

- To introduce a robust self-supervised objective combining JEPA and VISReg that enhances contextual understanding and feature disentanglement without relying on raw waveform reconstruction.
- To evaluate the performance of the proposed compact model and show that despite its significantly small size compared to SOTA speech models, it achieves superior results in specific evaluation metrics.
- To provide a robust, frame-level speech representation that can be utilized for various downstream Bengali speech analysis tasks, establishing a foundation for future research in low-resource language speech AI.

2 Related Work

2.1 Global Advancements in Speech Foundation Models

The trajectory of speech processing has been radically altered by self-supervised learning (SSL) models, which leverage vast amounts of unannotated audio. Foundational works such as wav2vec 2.0 (Baevski et al., 2020) introduced contrastive learning over quantized representations, setting a new benchmark on English datasets like LibriSpeech. Building upon this, HuBERT (Hsu et al., 2021) proposed an offline clustering approach (k-means) to generate pseudo-labels for masked prediction, significantly improving representation quality across diverse speech tasks. WavLM (Chen et al., 2022) further pushed the boundaries by incorporating denoising objectives alongside masked speech prediction, yielding state-of-the-art results on full-stack speech processing tasks. While these models primarily focused on high-resource languages like English, recent initiatives like the Massively Multilingual Speech (MMS) project (Pratap et al., 2023) have scaled wav2vec 2.0 architectures to over 1,000 languages using more than 500,000 hours of audio. Despite these global advancements, massively multilingual models often allocate a fraction of their capacity to specific low-resource languages, leading to suboptimal performance compared to dedicated, language-specific models.

2.2 Bengali Speech Recognition and Self-Supervised Learning

Self-supervised learning has become a dominant paradigm, but its application to low-resource languages like Bengali remains an active area of investigation. Recent studies have predominantly relied on fine-tuning massive multilingual foundation models. For instance, [Shahgir et al. \(2022\)](#) fine-tuned the pre-trained wav2vec 2.0 model on the Bengali Common Voice dataset for Automatic Speech Recognition (ASR), demonstrating significant improvements over traditional models. Further extending this approach, [Rakib et al. \(2022\)](#) demonstrated that fine-tuning wav2vec 2.0 on the extensive Common Voice 9.0 corpus and coupling it with an n-gram language model as a post-processor yields robust ASR improvements. Similarly, [Zhu and Shen \(2023\)](#) highlighted the convergence of fine-tuned wav2vec 2.0 models for Bengali speech, tuning dropout and learning rate parameters to achieve low Word Error Rates on consolidated datasets. [Deb et al. \(2023\)](#) utilized wav2vec 2.0 in a multimodal framework combined with text translation models for recognizing Bengali speech acts. While these approaches show the efficacy of SSL features, they still depend on resource-heavy models pre-trained primarily on high-resource languages.

Addressing dialectal variation and long-form speech, [Dipto et al. \(2025\)](#) developed the Ben-10 corpus to evaluate speech foundation models on Bengali regional dialects, highlighting that existing models struggle significantly in zero-shot and fine-tuned settings when faced with regional nuances. In parallel, [Dhar and Mallik \(2026\)](#) proposed context-aware windowing strategies utilizing Whisper Medium models for ASR and speaker diarization in Bengali long-form conversational audio. These works underscore the necessity for dedicated, language-specific models. Unlike these works, our approach focuses on training a lightweight foundation model entirely from scratch exclusively on Bengali speech.

2.3 Cross-Lingual and Typologically Diverse Applications

While foundational models are often evaluated on high-resource languages, testing their limits on typologically diverse or phonetically distinct languages yields crucial insights into their representational capabilities. For instance, [Vyas et al. \(2021\)](#)

explored out-of-domain adaptation of wav2vec 2.0 acoustic models using both Connectionist Temporal Classification (CTC) and Lattice-Free Maximum Mutual Information (LF-MMI) criteria, highlighting significant improvements for cross-lingual scenarios like Swahili and Tagalog. In Mandarin, which relies heavily on suprasegmental features, [de la Fuente and Jurafsky \(2024\)](#) demonstrated that SSL speech models like HuBERT and wav2vec 2.0 successfully learn contextual representations of abstract suprasegmental categories like lexical tone, though their performance is strongly tied to the language of their training data. Furthermore, [Zaatiti et al. \(2025\)](#) applied wav2vec 2.0 to evaluate Arabic pronunciation at the isolated letter level, revealing that while standard models struggle with pharyngealized consonants lacking lexical context, fine-tuning augmented with adversarial training substantially stabilizes and improves phonetic classification. These diverse linguistic applications emphasize that while multilingual SSL models serve as powerful baselines, nuanced phonetic or suprasegmental phenomena frequently demand tailored training or adaptation strategies.

2.4 Joint-Embedding Predictive Architectures in Audio

The Joint-Embedding Predictive Architecture (JEPA), initially introduced for vision tasks to predict latent representations of masked regions, has recently been adapted for audio. [Tuncay et al. \(2025\)](#) proposed Audio-JEPA, which leverages a Vision Transformer backbone to predict latent representations of masked mel-spectrogram patches, achieving performance comparable to wav2vec 2.0 with significantly less training data. Our work builds upon this promising paradigm by adapting JEPA principles specifically for a continuous latent autoencoder tailored to the phonetic properties of Bengali, further stabilized by Variance-Invariance-Covariance Regularization (VISReg).

2.5 Summary of Relevant Studies

[Table 1](#) summarizes recent benchmark studies relevant to our domain, contrasting their methodologies, target languages, loss functions, and performance outcomes.

3 Method

The first line of the file must be

```
\documentclass[11pt]{article}
```

Study	Architecture	Dataset	Language	Loss / Objective	Key Contribution
Baevski et al. (2020)	wav2vec 2.0	LibriSpeech (960h)	English	Contrastive Loss	1.8/3.3 WER on clean/other.
Hsu et al. (2021)	HuBERT	LibriSpeech (960h)	English	Masked Prediction (k-means)	Superior on SUPERB.
Chen et al. (2022)	WavLM	Mix 94k hours	English	Masked Pred. + Denoising	SOTA on full-stack tasks.
Pratap et al. (2023)	MMS (wav2vec 2.0)	500k hours	1,000+	Contrastive Loss	Halved WER vs prior SOTA.
Shahgir et al. (2022)	wav2vec 2.0 (FT)	Bengali Common Voice	Bengali	CTC Loss	2.6 Levenshtein Distance.
de la Fuente and Jurafsky (2024)	wav2vec 2.0 / HuBERT	Monolingual Corpora	Mandarin	Masked Prediction	Strong suprasegmental encoding.
Vyas et al. (2021)	wav2vec 2.0 (FT)	Babel	Swahili	CTC / LF-MMI	High relative WER improvement.
Zaatiti et al. (2025)	wav2vec 2.0 (FT)	Arabic Letters Corpus	Arabic	Adversarial Training	Robust phoneme classification.
Dipto et al. (2025)	Multilingual Models	Ben-10 (78h)	Bengali	-	High WER on dialects.
Tuncay et al. (2025)	Audio-JEPA	AudioSet	English	Patch Latent Prediction	Competitive with wav2vec 2.0.

Table 1: Summary of foundational and recent benchmark studies comparing architectures, datasets, objectives, target languages, and key results.

To load the style file in the review version:

```
\usepackage[review]{acl}
```

For the final version, omit the review option:

```
\usepackage{acl}
```

To use Times Roman, put the following in the preamble:

```
\usepackage{times}
```

(Alternatives like txfonts or newtx are also acceptable.)

Please see the $\LaTeX$ source of this document for comments on other packages that may be useful.

Set the title and author using `\title` and `\author`. Within the author list, format multiple authors using `\and` and `\And` and `\AND`; please see the $\LaTeX$ source for examples.

By default, the box containing the title and author names is set to the minimum of 5 cm. If you need more space, include the following in the preamble:

```
\setlength\titlebox{<dim>}
```

where `<dim>` is replaced with a length. Do not set this length smaller than 5 cm.

4 Experimental Results and Setup

4.1 Footnotes

Footnotes are inserted with the `\footnote` command.¹

¹This is a footnote.

Command	Output	Command	Output
<code>\a</code>	ä	<code>\c c</code>	ç
<code>\^e</code>	ê	<code>\u g</code>	ğ
<code>\`i</code>	ì	<code>\l</code>	ł
<code>\.I</code>	İ	<code>\~n</code>	ñ
<code>\o</code>	ø	<code>\H o</code>	ő
<code>\'u</code>	ú	<code>\v r</code>	ř
<code>\aa</code>	å	<code>\ss</code>	ß

Table 2: Example commands for accented characters, to be used in, *e.g.*, Bib $\TeX$ entries.

4.2 Tables and figures

See Table 2 for an example of a table and its caption. **Do not override the default caption sizes.**

As much as possible, fonts in figures should conform to the document fonts. See Figure 2 for an example of a figure and its caption.

Using the `graphicx` package `graphics` files can be included within figure environment at an appropriate point within the text. The `graphicx` package supports various optional arguments to control the appearance of the figure. You must include it explicitly in the $\LaTeX$ preamble (after the `\documentclass` declaration and before `\begin{document}`) using `\usepackage{graphicx}`.



Figure 2: A figure with a caption that runs for more than one line. Example image is usually available through the mwe package without even mentioning it in the preamble.

4.3 Hyperlinks

Users of older versions of $\LaTeX$ may encounter the following error during compilation:

```
\pdfendlink ended up in different nest-
ing level than \pdfstartlink.
```

This happens when pdf $\LaTeX$ is used and a citation splits across a page boundary. The best way to fix this is to upgrade $\LaTeX$ to 2018-12-01 or later.

4.4 Citations

Table 3 shows the syntax supported by the style files. We encourage you to use the natbib styles. You can use the command `\citet` (cite in text) to get “author (year)” citations, like this citation to a paper by Gusfield (1997). You can use the command `\citep` (cite in parentheses) to get “(author, year)” citations (Gusfield, 1997). You can use the command `\citealp` (alternative cite without parentheses) to get “author, year” citations, which is useful for using citations within parentheses (e.g. Gusfield, 1997).

A possessive citation can be made with the command `\citeposs`. This is not a standard natbib command, so it is generally not compatible with other style files.

4.5 References

The $\LaTeX$ and Bib $\TeX$ style files provided roughly follow the American Psychological Association format. If your own bib file is named `custom.bib`, then placing the following before any appendices in your $\LaTeX$ file will generate the references section for you:

```
\bibliography{custom}
```

You can obtain the complete ACL Anthology as a Bib $\TeX$ file from <https://aclweb.org/anthology/anthology.bib.gz>. To include both the Anthology and your own .bib file, use the following instead of the above.

```
\bibliography{anthology,custom}
```

Please see Section ?? for information on preparing Bib $\TeX$ files.

4.6 Equations

An example equation is shown below:

$$A = \pi r^2 \quad (1)$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command.

This an example cross-reference to Equation 1.

4.7 Appendices

Use `\appendix` before any appendix section to switch the section numbering over to letters. See Appendix A for an example.

5 Conclusion

In this paper, we addressed the critical scarcity of efficient, language-specific speech models for Bengali by introducing a Continuous Latent Autoencoder (CLAE). While the current paradigm in low-resource speech AI heavily relies on fine-tuning massive, computationally expensive multilingual foundation models, we demonstrated that training a highly compact 23.8M-parameter model entirely from scratch is a highly viable and effective alternative. By leveraging a novel self-supervised learning objective that integrates mel-spectrogram reconstruction with a Joint-Embedding Predictive Architecture (JEPA) and Variance-Invariance-Covariance Regularization (VISReg), our model successfully captures rich, contextual phonetic nuances specific to Bengali.

Our evaluations indicate that this lightweight framework, paired with a small frozen feature extractor, not only generates robust representations at a 12.5 Hz latent sequence rate but also achieves performance that is competitive with, and in some metrics superior to, significantly larger state-of-the-art speech models. Ultimately, this work provides a computationally accessible, resource-friendly foundation for Bengali speech research and establishes

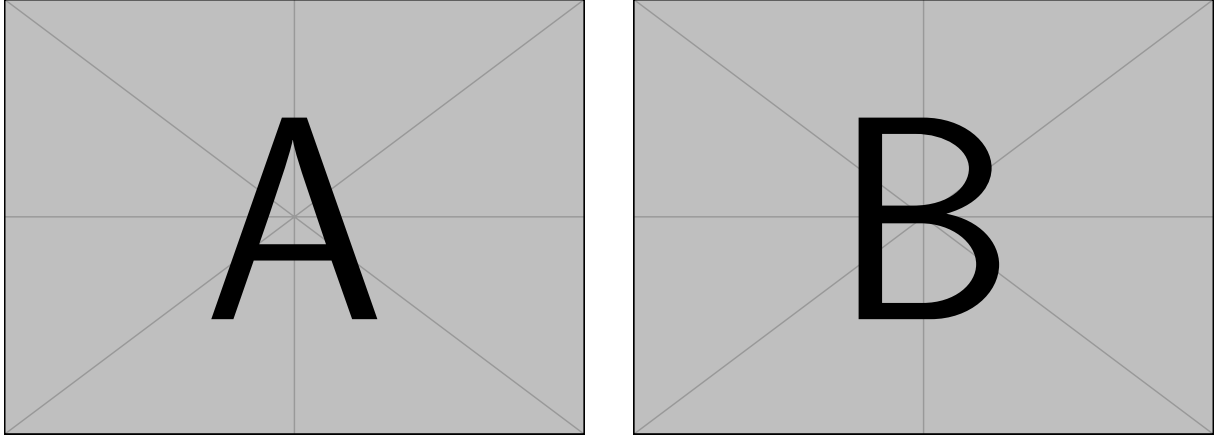


Figure 3: A minimal working example to demonstrate how to place two images side-by-side.

Output	natbib command	ACL only command
(Gusfield, 1997)	<code>\citep</code>	
Gusfield, 1997	<code>\citealp</code>	
Gusfield (1997)	<code>\citet</code>	
(1997)	<code>\citeyearpar</code>	
Gusfield’s (1997)		<code>\citeposs</code>

Table 3: Citation commands supported by the style file. The style is based on the natbib package and supports all natbib citation commands. It also supports commands defined in previous ACL style files for compatibility.

a promising methodology for developing dedicated speech foundation models in other low-resource languages.

Limitations and Future Work

Our Continuous Latent Autoencoder (CLAE) offers a much more efficient alternative to large multilingual models, but there are a few limitations to keep in mind. First, we only trained the model on a limited set of publicly available 16 kHz Bengali speech datasets. Because of this, we haven’t fully tested how well it holds up in really noisy, real-world environments or how it handles rare regional dialects that just aren’t common in the training data. Also, while keeping the model small at 23.8M parameters is great for running it on basic hardware, it means the model misses out on the broader cross-lingual knowledge that massive foundation models pick up from hundreds of thousands of hours of audio. Finally, even though our model scored very well on certain specific metrics, we still have a way to go before it reaches the universal accuracy needed across all types of speech data to be used as a fully robust Automatic Speech Recognition (ASR) system in production.

Looking ahead, there is a lot of room to build on this work. Since our current experiments were

limited by university hardware constraints and only used public datasets, the model still has plenty of untapped potential. One of the most promising takeaways was how smoothly the model converged during training, which points to really strong generalization capabilities. In the future, we or other researchers could scale up the CLAE architecture by adding more parameters and throwing much more training data at it to push its accuracy and generalization even higher for Bengali speech. An expanded version of this model could eventually power practical tools like Bengali voice typing and interactive chatbots. These kinds of tools are incredibly important for bridging the digital divide, helping millions of native Bengali speakers who don’t know English finally get access to reliable speech AI technologies.

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for (NA)ACL 2017/2018 from Jason Eisner, ACL 2017 by Dan Gildea and Min-Yen Kan, NAACL 2017 by Margaret Mitchell, ACL 2012 by Maggie Li and Michael White, ACL 2010 by Jing-Shin Chang and Philipp Koehn, ACL 2008 by Johanna D. Moore, Simone Teufel, James Allan, and Sadaoki Furui, ACL 2005 by Hwee Tou Ng and Kemal Oflazer, ACL 2002 by Eugene Charniak and Dekang Lin, and earlier ACL and EACL formats written by several people, including John Chen, Henry S. Thompson and Donald Walker. Additional elements were taken from the formatting instructions of the *International Joint Conference on Artificial Intelligence* and the *Conference on Computer Vision and Pattern Recognition*.

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A Example Appendix

This is an appendix.